

LIMON, CO, USA

WMO: 724665

Lat: 39.275N

Lon: 103.666W

Elev: 5374

StdP: 12.06

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93010

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -5.0	(c) 1.2	(d) -10.5	(e) 3.8	(f) 0.3	(g) -5.2	(h) 5.1	(i) 10.1	(j) 33.8	(k) 26.5	(l) 31.3	(m) 26.6	(n) 6.0	(o) 350	(p) 0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.5	93.4	60.8	90.9	60.8	88.3	60.5	66.1	81.0	64.9	80.3	63.8	79.7	11.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	103.2	68.8	60.9	97.7	68.2	59.2	91.9	67.5	33.7	80.7	32.7	80.1	31.8	79.1	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
29.6	25.7	22.2		-14.6	97.9	5.4	2.4	-18.5	99.6	-21.6	101.1	-24.7	102.4	-28.6	104.2	
				DB												
				WB	-15.3	68.9	5.4	1.4	-19.2	70.0	-22.4	70.8	-25.5	71.6	-29.4	72.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	27.2	28.4	37.5	44.7	54.5	65.2	71.3	68.9	60.7	47.4	35.8	26.4
	DBStd	17.95	10.14	10.50	9.48	8.52	7.89	6.14	4.08	4.42	6.94	8.94	9.68	10.39
	HDD50	3273	706	606	395	196	48	1	0	0	10	151	428	732
	HDD65	6863	1171	1025	853	610	333	71	7	17	157	547	875	1197
	CDD50	2337	0	1	7	36	186	457	659	587	332	69	3	0
	CDD65	453	0	0	0	0	7	77	201	139	29	0	0	0
	CDH74	8790	0	1	6	57	415	1706	3181	2177	1078	163	6	0
	CDH80	3758	0	0	0	4	105	731	1594	917	381	25	1	0
	WSAvg	9.4	8.2	9.4	10.4	12.1	10.8	9.9	8.8	8.4	8.6	9.2	8.5	8.5
	PrecAvg	16.2	0.4	0.4	0.7	1.4	2.2	2.4	2.9	2.8	1.1	0.9	0.5	0.2
Precipitation	PrecMax	20.2	0.9	1.1	1.9	5.3	4.5	4.7	7.1	4.1	3.1	2.4	1.4	0.7
	PrecMin	13.4	0.0	0.0	0.1	0.1	0.5	1.1	0.6	1.3	0.2	0.0	0.0	0.0
	PrecStd	1.9	0.3	0.3	0.6	1.1	1.3	1.2	1.5	1.0	0.8	0.8	0.4	0.2
	PrecStd	1.9	0.3	0.3	0.6	1.1	1.3	1.2	1.5	1.0	0.8	0.8	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.5	67.4	74.2	79.5	87.4	96.3	97.5	94.4	91.0	83.1	74.1	63.3
		MCWB	43.0	44.0	47.3	50.1	57.0	58.5	62.1	61.0	59.0	53.6	47.1	42.8
	2%	DB	57.8	60.6	69.9	75.0	82.4	91.4	94.5	91.2	88.0	78.9	69.2	57.0
		MCWB	39.6	41.0	45.5	49.3	55.2	59.3	61.2	60.9	57.7	52.2	45.5	39.5
	5%	DB	52.1	55.1	64.9	71.4	79.0	88.1	91.8	88.7	84.5	74.1	63.1	51.7
		MCWB	37.6	39.0	43.8	47.8	54.5	59.0	61.6	61.1	57.2	50.3	43.9	37.4
	10%	DB	45.7	48.4	59.3	66.0	74.4	84.1	89.5	85.6	81.1	68.6	56.5	45.5
		MCWB	34.7	36.0	41.7	46.5	53.6	59.2	61.7	61.2	56.8	48.8	40.8	34.5
	0.4%	WB	43.9	44.7	49.4	53.7	61.9	66.6	68.1	67.8	63.8	57.1	49.1	43.5
		MCDB	62.6	64.0	68.9	69.7	78.8	80.0	82.2	81.7	78.9	73.0	68.4	61.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.5	41.9	46.9	51.2	59.0	64.4	66.3	65.9	61.7	54.5	46.4	40.3
		MCDB	56.2	58.9	65.8	68.9	73.9	79.3	81.5	80.8	76.8	71.2	65.9	55.8
	5%	WB	37.9	39.3	44.7	49.4	57.0	62.7	65.2	64.5	60.2	52.3	44.3	37.5
		MCDB	51.0	53.8	62.9	67.3	71.9	78.4	81.0	79.5	75.5	68.9	62.3	50.5
	10%	WB	34.9	36.5	42.3	47.6	55.1	61.2	64.2	63.4	58.8	50.1	41.3	34.6
		MCDB	45.2	48.2	58.2	64.0	69.9	77.3	80.6	78.3	74.2	66.1	55.8	45.1
Mean Daily Temperature Range	5% DB	MDBR	31.3	30.0	32.3	30.5	29.9	31.5	32.5	30.4	33.1	32.6	32.9	30.9
		MCDBR	40.2	41.0	42.7	42.8	39.2	39.6	39.3	36.7	40.5	43.5	43.8	40.5
		MCWBR	24.1	23.6	21.9	20.1	16.1	13.7	12.4	12.0	15.2	20.1	23.4	24.6
	5% WB	MCDBR	38.3	38.1	38.9	37.4	32.8	32.5	30.8	29.7	33.1	36.0	41.6	38.8
		MCWBR	23.7	22.6	20.7	18.4	15.1	13.4	11.3	11.5	13.7	17.9	22.8	24.2
Clear-Sky Solar Irradiance	taub		0.235	0.243	0.265	0.292	0.316	0.329	0.352	0.343	0.306	0.267	0.248	0.237
	taud		2.614	2.607	2.541	2.476	2.469	2.450	2.435	2.451	2.523	2.620	2.589	2.591
	Ebn at Noon		309	319	318	313	305	299	292	292	299	305	300	300
	Edh at Noon		21	25	30	34	35	36	36	34	30	24	22	20
All-Sky Solar Radiation	RadAvg		826	1101	1512	1859	2112	2377	2327	2082	1722	1260	906	722
	RadStd		44	69	109	105	143	101	88	60	83	82	57	40

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	N/A

Nomenclature: See separate page